

Electric Circuits Lab

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Resistance Measurement

March 9, 2025

Overview

Here is book of Electric Circuits Lab experiment on page 47

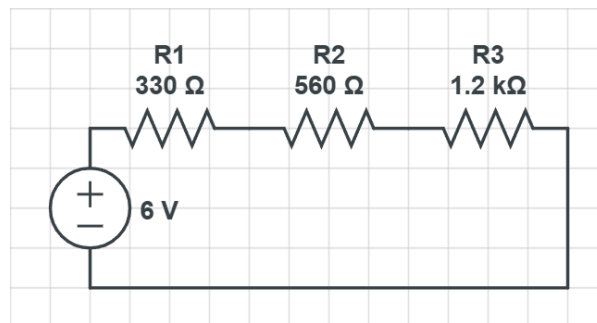
Experiment Info

- **Purpose** > The purpose of this experiment is to determine the resistance values of 3 resistors using their color codes, measure their actual resistance using an ohmmeter, and compare the theoretical and measured values to calculate the percentage error.
- **Necessary Equipment** > Resistors, DC Power, Multimeter, Wires

Ohm's Law : $V = I \cdot R$

$$V = V_s \left(\frac{R}{R_1 + R_2 + R_3} \right)$$

$$I_{total} = I_1 + I_2 + I_3$$



KVL: $-6 + 320I + 560I + 1200I = 0 \Rightarrow I = 2.871 \text{ mA}$

Steps

1. First we picked resistors by their color code.
2. Then we put resistors in series on the breadboard.
3. Then turned off stand-by mode and measured the real resistance of each resistor.

Resistor Color and Voltage Table

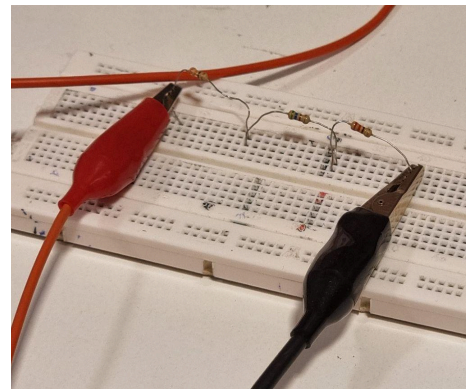
Resistor	Color 1	Color 2	Color 3	V_{real}	$V_{measured}$
$R_1 = 330\Omega$	Orange (3)	Orange (3)	Brown (1)	0.95V	0.97V
$R_2 = 560\Omega$	Green (5)	Blue (6)	Brown (1)	1.61V	1.62V
$R_3 = 1200\Omega$	Brown (1)	Red (2)	Red (2)	3.45V	3.50V

Calculations of V_{real} :

$$V_{1-real} = I_1 \times R_1 = 0.002871 \times 330 \approx 0.95 V$$

$$V_{2-real} = I_1 \times R_1 = 0.002871 \times 560 \approx 1.61 V$$

$$V_{2-real} = I_1 \times R_1 = 0.002871 \times 1200 \approx 3.45 V$$



Analyzing R-L-C Circuit - Error Calculations

Error calculations are divided into 3 categories, absolute, relative and systematic errors.

from absolute formula :

$$\text{Absolute Error} = |\text{real value} - \text{measured value}|$$

We get this absolute error for α :

$$\varepsilon_{abs_{V_{1-real}}} = |0.95V - 0.97V| = 0.02V \quad \varepsilon_{abs_{V_{2-real}}} = |1.61V - 1.62V| = 0.01V$$

$$\varepsilon_{abs_{V_{3-real}}} = |3.45V - 3.50V| = 0.05V$$

from relative formula :

$$\text{Relative Error} = \frac{|\text{real value} - \text{measured value}|}{\text{real value}}$$

We get this absolute error for V_{1-real} :

$$\varepsilon_{rel_{V_{1-real}}} = \frac{0.02V}{0.95V} \approx 0.02105$$

We get this absolute error for V_{2-real} :

$$\varepsilon_{rel_{V_{2-real}}} = \frac{0.01V}{1.61V} \approx 0.00621$$

We get this absolute error for V_{3-real} :

$$\varepsilon_{rel_{V_{3-real}}} = \frac{0.05V}{3.45V} \approx 0.01449$$

for systematic formula, We must do several mathematics operations :

1. Consider we have formula $V = I \cdot R$
2. Take the $\ln$ from both sides.
3. Take the differential from both sides.
4. Convert differentials to delta (Δ).
5. Convert every $-$ to $+$.
6. accuracy of **Analog Device** is equal to the lowest shown range on the device.
7. Error of **Analog Device** is equal to the lowest shown range on device multiplied by 0.5.
8. accuracy of **Digital Device** is equal to 10^{-x} where x is float level on the screen.
9. Error of **Digital Device** is equal to 10^{-x} where x is float level on the screen.

Now we can calculate systematic error using main formula :

$$\text{Step 1: } V = I \cdot R$$

$$\text{Step 2: } \ln(V) = \ln(I) + \ln(R)$$

$$\text{Step 3: } \frac{dV_n}{V_n} = \frac{d(I_{total})}{I_{total}} + \frac{d(R_n)}{R_n}$$

$$\text{Step 4: } \frac{dV_n}{V_n} = \frac{\Delta I_{total}}{I_{total}} + \frac{\Delta R_n}{R_n}$$

$$\text{Step 6~: } \Delta I = 10^{-3} , \quad \Delta R = 10^{-3}$$

$$\text{for } V_1 : \quad \varepsilon_{sys_{V_1}} = \frac{\Delta V_1}{V_1} = \frac{0.001}{0.002871} + \frac{0.001}{0.330} \approx 0.3513$$

$$\text{for } V_2 : \quad \varepsilon_{sys_{V_2}} = \frac{\Delta V_2}{V_2} = \frac{0.001}{0.002871} + \frac{0.001}{0.560} \approx 0.3501$$

$$\text{for } V_3 : \quad \varepsilon_{sys_{V_2}} = \frac{\Delta V_3}{V_3} = \frac{0.001}{0.002871} + \frac{0.001}{1200} \approx 0.3491$$